

# Factorial And Transfer Protocol

This document defines the follow-on study after the first adversarial-curriculum milestone. It has two linked parts:

1. Phase 2b: a holdout-first factorial ablation in the original resource-collection environment.
2. Phase 3: cross-environment transfer of the best curriculum recipe.

For the next step after the initial transfer result, use docs/PHASE\_4\_CAUSAL\_TRANSFER\_PROTOCOL.md. For the benchmark-transfer step after the causal interpretation phase, start with docs/TSP\_PHASE\_5\_PROTOCOL.md and then use docs/ROUTING\_PHASE\_5B\_5C\_PROTOCOL.md for the ATSP/CVRP extensions.

## Why This Exists

The earlier curriculum suite established that adversarial pressure can sustain adaptation, and that opponent diversity plus memory look more promising than simple mutation forcing. It did not yet settle three stronger questions:

1. Which curriculum ingredients actually improve held-out generalization?
2. Are the largest novelty spikes real strategy changes or just code churn?
3. Do the same curriculum principles hold outside the original resource-collection benchmark?

This protocol addresses those gaps directly.

## Research Basis

The design follows standard best practices from reproducible reinforcement learning, quality-diversity search, and generalization-focused evaluation:

- Deep Reinforcement Learning That Matters (<https://arxiv.org/abs/1709.06560>): replicated runs, uncertainty, and evaluation discipline.
- A Study on Overfitting in Deep Reinforcement Learning (<https://research.google/pubs/a-study-on-overfitting-in-deep-reinforcement-learning/>): train/test separation and the risk of narrow benchmark overfitting.
- Novelty Search and the Problem with Objectives ([https://doi.org/10.1007/978-1-4614-1770-5\\_3](https://doi.org/10.1007/978-1-4614-1770-5_3)): novelty as a useful search signal, but not equivalent to task performance.
- Illuminating search spaces by mapping elites (<https://arxiv.org/abs/1504.04909>): behavior-space coverage and quality-diversity archives.
- Open-ended Learning in Symmetric Zero-sum Games (<https://proceedings.mlr.press/v97/balduzzi19a.html>): opponent diversity, non-transitive adaptation, and the role of memory in open-ended learning.
- Measuring Sample Efficiency and Generalization in RL with Procggen (<https://openreview.net/forum?id=rylKB3A9Fm>): transfer and out-of-distribution evaluation matter when making general claims.

## Phase 2b: Holdout-First Factorial Ablation

### Goal

Compare six curriculum recipes while using held-out opponent win rate as the primary endpoint:

1. fixed predator
2. rotating opponents

3. rotating + nemesis archive
4. rotating + novelty gate
5. rotating + replay-aware selection
6. rotating + nemesis + novelty + replay

## Endpoint Hierarchy

1. Primary: held-out opponent win rate
2. Secondary: held-out score margin
3. Supporting context: training score, curriculum metrics, novelty, and reliability

Training score alone is not sufficient evidence of generalizable adaptation.

## Replication

- Use five seed offsets, giving 30 condition-runs total.
- Aggregate only after all replicates finish.
- If one replicate is contaminated by provider failures, rerun only that replicate.

## Clean Interpretation Rules

- Do not rank recipes by training score if the holdout endpoint disagrees.
- Do not treat code novelty as innovation without the novelty-review packet.
- Treat replay checks as training filters, not final evidence.
- Keep the holdout panel separate from training-time opponent pools.

## Novelty Validation Packet

The factorial study must be followed by manual validation of the highest novelty spikes.

For each selected epoch, inspect:

1. code diff
2. behavioral descriptor change
3. trajectory change
4. score change
5. holdout-effect change, when available

This step exists to answer the strongest reviewer objection: whether “novelty” is real strategic change or mostly code churn.

The packet is generated by `review_novelty_spikes.py`.

## Phase 3: Cross-Environment Transfer

### Goal

Take the best factorial recipe and test whether its curriculum principles transfer to additional simple games:

1. resource collection / denial
2. pursuit / evasion
3. territory control

## **Interpretation**

The question is not whether absolute scores are identical across games. The question is whether the same curriculum recipe continues to improve held-out win rate under different objective structures.

## **Transfer Rules**

- Choose the winning recipe from the factorial aggregate first.
- Generate the transfer suite from that recipe.
- Run five replicate seed offsets again.
- Keep environment-specific opponent pools and holdout panels distinct.
- Report results per environment before making any cross-environment summary.

The transfer-suite generator is `build_transfer_suite.py`.

## **Required Artifacts**

- factorial aggregate report
- novelty-review packet
- transfer aggregate report
- qualitative examples from both the strongest and weakest transfer conditions

## **Decision Rule For Next Iteration**

If the factorial winner improves holdout win rate but does not transfer, the result is still valuable as an environment-specific curriculum finding.

If the factorial winner also transfers across the added games, then the project can support a stronger methods claim about curriculum pressure for studying LLM-agent adaptation, rather than only a toy-grid result.

If the transfer signal is promising but the mechanism remains unclear, the next phase is not another mechanism search. The next phase is paired replication, failure-mode profiling, and behavior-versus-novelty interpretation under `docs/PHASE_4_CAUSAL_TRANSFER_PROTOCOL.md`.